Aman Verma

SOFTWARE DEVELOPER

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WORK EXPERIENCE

Software Developer Intern Jan 2026 - Jun 2026

LexoLegal *Remote*

- Architected asynchronous **REST APIs** in **FastAPI**, **SQLAlchemy 2.0**, and **PostgreSQL** (SQL) behind an Nginx reverse proxy, implementing multi-tier JWT authentication, RFC 9457 standardized problem details, and Alembic migrations with **Postman** test suites.
- Engineered background worker pipelines using **Celery** with atomic **Redis** distributed locks and non-blocking 10s countdown requeues, eliminating race conditions from rapid webhook events.
- Designed an automated conversational assistant utilizing Pydantic AI and webhooks with HMAC-SHA256 verification, processing inquiries with **sub-2s latency** and zero prompt leakage.
- Automated daily judicial data ingestion across 25 High Courts and 743 District Courts via **Celery** workers in **Python** and **Bash**, using Selectolax DOM extractors and **PostgreSQL** batch inserts.

Freelance Developer Oct 2024 - Jan 2025

Webxela *Remote*

- Implemented Square POS **OAuth 2.0** authentication and webhook ingestion pipelines in **Spring Boot** (**Java**) and Spring Data JPA, automating merchant onboarding with idempotent transaction verification.
- Developed a concurrent search and scraping service in **Kotlin**, **Spring Boot**, and **PostgreSQL**, utilizing Coroutines with rate-limited HTTP connections and indexed trigram search queries verified with **JUnit 5**.

PROJECTS

doubtless | *Python, FastAPI, Celery, Redis, SQLite, Docker, OpenTelemetry, Logfire, Git* GitHub

- Streamlined real-time solutions for academic queries by developing an interactive assistant in **FastAPI**, streaming solutions with live context via Server-Sent Events (SSE).
- Constructed an asynchronous 4-stage media ingestion pipeline (PipelineRunner) in **Celery** and **Redis** that transcodes HLS streams, slices audio (<15MB/hr), and transcribes lectures in <10s.
- Instrumented distributed tracing across services via **OpenTelemetry** and **Logfire**; built the async data layer with thread-safe **SQLite** WAL repositories and **Docker** containers (151 tests in **Pytest** and **Git** CI).

watchtower | *Kotlin, Room, WorkManager, OkHttp, Clean Architecture, DSA* GitHub

- Created a local-first web monitoring app utilizing MD5 checksum hashing and Myers diff algorithm (**DSA**) to track DOM modifications and dispatch push alerts.
- Optimized concurrent network inspection via **Kotlin** Coroutines with semaphore rate limiting, executing background jobs via **WorkManager** while maintaining a consistent **60 FPS** UI.
- Structured persistent offline storage in **Clean Architecture** using **Room** database with Flow observables and OkHttp client with custom SSL trust managers.

ncert-rag-evals | *Python, SQLite, Pytest, GitHub Actions* GitHub

- Evaluated empirical retrieval benchmarks across 17 STEM textbooks in **Python**, implementing custom hybrid search in **SQLite** FTS5 to profile retrieval strategies on raw CPU latency.
- Benchmarked Reciprocal Rank Fusion against BM25 and dense embeddings, achieving **86.9% Recall@5 in 8.9ms on CPU** while cutting retrieval latency by **99.8%**.
- Formulated a section-aware parser indexing 4,429 chunks into **SQLite**; verified statistical significance via paired McNemar tests (**p = 0.0021**) across 92 unit tests in **Pytest** with **GitHub Actions**.

TECHNICAL SKILLS

Languages & Core: Python, Kotlin, Java, SQL, Bash, DSA

Backend & Data Systems: FastAPI, Spring Boot, PostgreSQL, Redis, Celery, REST APIs, SQLite, Docker

Architecture & Design: Clean Architecture, Room, WorkManager, OAuth 2.0

Testing & Developer Tools: Git, GitHub Actions, Pytest, JUnit 5, Postman, OpenTelemetry, Logfire

LEADERSHIP & ACTIVITIES

- Delivered 3 technical sessions as a speaker on 'Generative AI Unlocked: LLM Basics' and developer tooling, training 150+ students on system architecture, prompt engineering, and API integration.
- Coordinated placement drives and technical interview logistics at CTPC (Central Training & Placement Cell), organizing interview scheduling and operational workflows for 100+ candidates.

EDUCATION

Veer Bahadur Singh Purvanchal University 2023 - 2027

Bachelor of Technology in Computer Science & Engineering *Jaunpur, India*